

12 Minimum field data package

A consistent field record turns a nutrient flag into a testable diagnosis and makes later model updates more useful.

Area	Minimum record
Leaf sample	Cultivar rootstock block date leaf age and position crop load symptoms spray residue laboratory and method
Irrigation water	EC pH bicarbonate sodium chloride boron calcium magnesium sulfate nitrate and SAR when relevant
Soil	Depth position in wetted pattern texture pH ECe organic matter CEC exchangeable cations available nutrients SAR or sodium measure
Irrigation operation	System output and uniformity frequency duration wetted width and depth seasonal applied water runoff ponding and repair history
Root and soil condition	Feeder-root color density depth Phytophthora history compaction restrictive layers infiltration drainage aeration and mulch depth
Inputs	Product analysis source rate date placement treated area and all fertilizer amendment compost manure mulch acid and microbial products
Weather and crop	Heat cold rainfall wind harvest yield fruit size alternate-bearing stage and major canopy or root events
Response	Comparable untreated area photos new growth follow-up leaf soil water and root measures yield quality and cost

Sampling practices that prevent false conclusions

- **Use the same protocol over time.** Comparable leaf age, position, season, laboratory, and method are essential for trend interpretation.
- **Sample meaningful zones separately.** Do not blend healthy and affected areas, wet and dry emitter zones, or shallow and deep layers when the difference is the question.
- **Record the water context.** A soil or leaf sample without recent irrigation, rainfall, salinity, and crop-water information can be easy to misread.
- **Verify the intervention.** Confirm where the material and water actually went, then measure roots and new growth before claiming success.

Resource bank role These appendices are designed to be attached to the main DST report or broken into shorter advisory cards. Local numeric recommendations should be added only after regionally appropriate calibration and review.